Dissipative Bounds and Ruiz Castillo Residual Decomposition in the Accelerated Dynamics of the Collatz Conjecture

ScD. Juan Carlos Ruiz Castillo

University of San Carlos of Guatemala

jcefpem@profesor.usac.edu.gt

DOI: https://doi.org/10.5281/zenodo.21184721

2026

Abstract

The present work develops a new advance within the theory of residual debt and residual drift applied to the accelerated dynamics of the Collatz Conjecture. Starting from the accelerated map

$$U(n) = \frac{3n+1}{2^{\nu_2(3n+1)}},$$

the exact affine representation

$$U^k(n) = \frac{3^k n + B_k(n)}{2^{A_k(n)}}$$

is studied, where $A_k(n)$ represents the accumulated binary dissipation and $B_k(n)$ collects the arithmetic memory induced by the additive terms $+1$. The *Ruiz Castillo residual decomposition* is formally introduced,

$$U^k(n) = 2^{L_k(n)} n + R_k(n),$$

where

$$L_k(n) = k \log_2(3) - A_k(n)$$

is the accumulated residual debt and

$$R_k(n) = \frac{B_k(n)}{2^{A_k(n)}}$$

is the normalized residue. This decomposition separates the accelerated dynamics into two components: a dominant multiplicative term, controlled by the residual debt, and a normalized residual memory term, controlled by the complete structure of the 2-adic valuations. The article proves the symbolic formula for $R_k(n)$, obtains universal bounds and bounds under hypotheses of average tail dissipation, and introduces a residual criterion for effective descent. The objective is not to assert a final proof of the Collatz Conjecture, but rather to construct a mathematical architecture for studying the relationship among negative drift, dissipative pressure, residual memory, and effective descent of accelerated orbits.

Keywords: Collatz Conjecture; residual debt; residual drift; Ruiz Castillo residual decomposition; normalized residue; 2-adic valuation; dissipative pressure; discrete dynamical systems.

1 Introduction

The Collatz Conjecture constitutes one of the best-known, most persistent, and most suggestive open problems in contemporary number theory. Its elementary formulation contrasts notably with the arithmetic, dynamical, and computational complexity that emerges when studying its orbits. For each positive integer n, the classical Collatz operator is defined by

$$C(n) = \begin{cases} \dfrac{n}{2}, & n \equiv 0 \pmod 2, \\ 3n+1, & n \equiv 1 \pmod 2. \end{cases}$$

The conjecture states that, for every $n \in \mathbb{N}$, there exists an integer $r \geq 0$ such that

$$C^r(n) = 1.$$

Once the value 1 has been reached, the orbit enters the trivial cycle

$$1 \longrightarrow 4 \longrightarrow 2 \longrightarrow 1.$$

Consequently, the conjecture can be equivalently formulated as the assertion that there are no divergent positive orbits and no positive cycles distinct from the trivial cycle.

The difficulty of the problem does not lie in the definition of the operator, but in the interaction between two opposing arithmetic mechanisms. On the one hand, when n is odd, the transformation

$$n \longmapsto 3n+1$$

introduces an expansion of ternary nature. On the other hand, whenever an even number appears, the successive divisions by powers of 2 introduce a mechanism of reduction or binary dissipation. Thus, Collatz dynamics can be interpreted as an accumulative competition between growth and dissipation:

$$\text{ternary growth} \qquad \text{versus} \qquad \text{binary dissipation.}$$

From this perspective, the problem does not consist only in observing whether a particular orbit increases or decreases in certain isolated steps. The deeper question is to determine whether, along every positive orbit, the accumulated binary dissipation always eventually compensates for the ternary growth generated by the odd steps.

A natural strategy for studying this tension is to restrict the dynamics to the odd integers and to consider the accelerated Collatz map. Denote by

$$2\mathbb{N} + 1 = 2\mathbb{N} + 1$$

the set of positive odd integers. For $n \in 2\mathbb{N} + 1$, define

$$U(n) = \frac{3n+1}{2^{\nu_2(3n+1)}}, \qquad n \in 2\mathbb{N} + 1,$$

where $\nu_2(m)$ represents the largest power of 2 that divides m. The map U eliminates the intermediate even steps of the classical operator and records only the succession of odd values that appear in the Collatz orbit.

In this way, each iteration of U concentrates two operations in a single step: first, the growth produced by $3n + 1$; second, the dissipation produced by dividing by the largest possible power of 2. This formulation allows the dynamics to be studied through a sequence of odd integers

$$n_0 = n, \qquad n_{j+1} = U(n_j),$$

or, equivalently,

$$n_j = U^j(n).$$

A 2-adic valuation is associated with each accelerated step:

$$a_j(n) = \nu_2(3U^j(n) + 1).$$

The sequence

$$(a_0(n), a_1(n), a_2(n), \ldots)$$

can be interpreted as the dissipative word of the orbit. Each term $a_j(n)$ measures how much binary dissipation occurs after the corresponding expansive step.

For a block of length k, the accumulated binary dissipation is defined by

$$A_k(n) = \sum_{j=0}^{k-1} a_j(n).$$

This quantity records the total number of divisions by 2 that occur during the first k accelerated steps. On the other hand, after k odd steps, the dominant ternary growth is governed by the factor

$$3^k.$$

Expressing this growth in binary scale gives

$$3^k = 2^{k \log_2(3)}.$$

Therefore, the value

$$k \log_2(3)$$

represents the critical threshold of binary dissipation needed to compensate for the dominant ternary growth after k steps.

Along these lines, the residual debt is introduced:

$$L_k(n) = k \log_2(3) - A_k(n).$$

The quantity $L_k(n)$ measures, on a binary logarithmic scale, the deficit or surplus between accumulated ternary growth and accumulated binary dissipation. Indeed,

$$L_k(n) = \log_2 \left(\frac{3^k}{2^{A_k(n)}} \right).$$

Thus, if

$$L_k(n) > 0,$$

the block preserves expansive debt, because the accumulated dissipation has not completely compensated for ternary growth. In contrast, if

$$L_k(n) < 0,$$

the block possesses dissipative surplus, since binary dissipation exceeds the threshold required to neutralize the dominant growth.

This formulation allows a dynamical reading of the residual debt to be introduced. If the residual increment is defined by

$$X_j(n) = \log_2(3) - a_j(n),$$

then the accumulative identity

$$L_k(n) = \sum_{j=0}^{k-1} X_j(n)$$

is obtained. Therefore, the residual debt can be interpreted as a symbolic walk governed by the 2-adic valuations. In this framework, the residual drift

$$D_k(n) = \frac{L_k(n)}{k}$$

measures the average tendency of that walk. A negative value of $D_k(n)$ indicates mean dissipative pressure; a positive value indicates mean debt accumulation.

However, although negative residual debt is a relevant condition, it is not by itself sufficient to guarantee effective descent of the orbit. The fundamental reason is that the accelerated iteration is not determined only by the dominant factor

$$\frac{3^k}{2^{A_k(n)}}.$$

Each odd step contains an additive term $+1$, and these terms accumulate throughout the iteration. This accumulation produces an arithmetic residue that cannot be ignored.

Indeed, the accelerated iteration admits an exact affine representation:

$$U^k(n) = \frac{3^k n + B_k(n)}{2^{A_k(n)}}.$$

Here, $B_k(n)$ represents the accumulated memory of the additive terms $+1$. This expression shows that the behavior of $U^k(n)$ depends on two elements: the dominant quotient

$$\frac{3^k}{2^{A_k(n)}}$$

and the normalized residual term

$$\frac{B_k(n)}{2^{A_k(n)}}.$$

The present work develops a new stage of this theory by introducing the normalized residue

$$R_k(n) = \frac{B_k(n)}{2^{A_k(n)}}.$$

With this definition, the exact affine form becomes the identity

$$U^k(n) = 2^{L_k(n)} n + R_k(n).$$

This expression will be called the *Ruiz Castillo residual decomposition*.

Central idea of the article

The *Ruiz Castillo residual decomposition*

$$U^k(n) = 2^{L_k(n)} n + R_k(n)$$

separates the accelerated dynamics into two fundamental components. The term

$$2^{L_k(n)} n$$

represents the dominant force of expansion or contraction, governed by the residual debt $L_k(n)$. The term

$$R_k(n)$$

represents the normalized residual memory, generated by the accumulation of the $+1$ terms in successive iterations.

This separation makes it possible to formulate the problem of effective descent not only as a question of negative drift, but as an interaction between dissipative pressure and control of the accumulated residue. Consequently, proving that an orbit has negative residual debt is not enough: it is also necessary to show that the normalized residue remains below the available contraction margin.

The objective of this article is not to present a final proof of the Collatz Conjecture, but to construct a finer mathematical architecture for studying the accelerated dynamics. In particular, it seeks to show that the analysis must be organized at two levels: first, the study of residual debt and dissipative drift; second, the control of the normalized residue $R_k(n)$, which constitutes the principal technical obstacle to transforming average dissipative pressure into effective descent of the orbits.

The central contribution of the work therefore consists in introducing and developing the decomposition

$$U^k(n) = 2^{L_k(n)} n + R_k(n),$$

together with dissipative bounds for $R_k(n)$. This formulation allows the accelerated dynamics to be reinterpreted as a system with dominant contraction and residual memory, opening a research path centered on the relationship among 2-adic valuations, residual drift, dissipative pressure, and stability of the orbits.

2 Accelerated map, valuations, and residual debt

The direct analysis of the classical Collatz operator presents an immediate structural difficulty: the orbit alternates between even and odd terms, and the number of successive divisions by 2 that appear after each odd step is not constant. To isolate the essential mechanism of growth and dissipation, it is convenient to concentrate the dynamics on the subset of odd integers.

Let

$$2\mathbb{N} + 1 = 2\mathbb{N} + 1$$

be the set of positive odd integers. For every positive integer m, denote by $\nu_2(m)$ the 2-adic valuation of m, that is, the largest integer $\alpha \geq 0$ such that

$$2^{\alpha} \mid m.$$

Equivalently, there exists a unique odd integer q such that

$$m = 2^{\nu_2(m)} q.$$

Definition 2.1 (Accelerated Collatz map)**.** The accelerated Collatz map is defined by

$$U : 2\mathbb{N} + 1 \to 2\mathbb{N} + 1, \qquad U(n) = \frac{3n + 1}{2^{\nu_2(3n+1)}}.$$

Proposition 2.2 (Well-definedness of the accelerated map)**.** *For every* $n \in 2\mathbb{N} + 1$*, one has* $U(n) \in 2\mathbb{N} + 1$*. Therefore, the map*

$$U : 2\mathbb{N} + 1 \to 2\mathbb{N} + 1$$

is well defined.

Proof. Let $n \in 2\mathbb{N} + 1$. Then n is odd, so $3n$ is also odd. Consequently,

$$3n + 1$$

is even. It follows that

$$\nu_2(3n+1) \geq 1.$$

By definition of the 2-adic valuation, there exists a unique odd integer q such that

$$3n+1 = 2^{\nu_2(3n+1)}q.$$

Dividing by $2^{\nu_2(3n+1)}$, one obtains

$$q = \frac{3n+1}{2^{\nu_2(3n+1)}} = U(n).$$

Since q is odd, it follows that $U(n) \in 2\mathbb{N}+1$. □

Observation 2.3. The accelerated map does not eliminate essential information from the classical orbit. What it does is condense into a single step the odd transformation $3n+1$ together with all subsequent binary divisions. Thus, each accelerated iteration represents a complete dynamical unit: ternary expansion followed by maximal binary dissipation.

2.1 Accelerated orbit and valuation word

Let $n \in 2\mathbb{N}+1$. The accelerated orbit of n is defined by

$$n_0 = n, \qquad n_{j+1} = U(n_j), \qquad j \geq 0.$$

Equivalently,

$$n_j = U^j(n).$$

Definition 2.4 (Valuation word)**.** Let $n \in 2\mathbb{N}+1$. The valuation word associated with the accelerated orbit of n is defined by

$$\mathbf{a}(n) = (a_0(n), a_1(n), a_2(n), \ldots),$$

where

$$a_j(n) = \nu_2(3U^j(n)+1).$$

Each term $a_j(n)$ measures the intensity of the binary dissipation that appears after the corresponding expansive step. Indeed, if

$$n_j = U^j(n),$$

then

$$n_{j+1} = \frac{3n_j + 1}{2^{a_j(n)}}.$$

Therefore,

$$3n_j + 1 = 2^{a_j(n)} n_{j+1}.$$

Proposition 2.5 (Elementary transition equation)**.** *For every $n \in 2\mathbb{N} + 1$ and every $j \geq 0$, one has*

$$3U^j(n) + 1 = 2^{a_j(n)} U^{j+1}(n).$$

Proof. By definition,

$$a_j(n) = \nu_2(3U^j(n) + 1).$$

Then there exists a unique odd integer q such that

$$3U^j(n) + 1 = 2^{a_j(n)} q.$$

But, by definition of the accelerated map,

$$U^{j+1}(n) = U(U^j(n)) = \frac{3U^j(n) + 1}{2^{a_j(n)}}.$$

Therefore, $q = U^{j+1}(n)$, and one obtains

$$3U^j(n) + 1 = 2^{a_j(n)} U^{j+1}(n).$$

□

Symbolic reading

The accelerated orbit can be studied through two complementary representations:

$$n,\ U(n),\ U^2(n), \ldots$$

and

$$a_0(n),\ a_1(n),\ a_2(n), \ldots.$$

The first shows the numerical evolution of the orbit; the second shows the dissipative structure that governs that evolution.

2.2 Accumulated binary dissipation

Definition 2.6 (Accumulated binary dissipation)**.** For $k \geq 1$, define

$$A_k(n) = \sum_{j=0}^{k-1} a_j(n),$$

with the convention

$$A_0(n) = 0.$$

The quantity $A_k(n)$ records the total number of factors 2 removed during the first k accelerated steps. Since each $a_j(n) \geq 1$, the elementary bound

$$A_k(n) \geq k$$

holds.

Proposition 2.7 (Monotonicity of accumulated dissipation)**.** *For every* $n \in 2\mathbb{N}+1$ *and every* $k \geq 0$*,*

$$A_{k+1}(n) = A_k(n) + a_k(n).$$

In particular, the sequence $\{A_k(n)\}_{k \geq 0}$ *is strictly increasing.*

Proof. By definition,

$$A_{k+1}(n) = \sum_{j=0}^{k} a_j(n).$$

Separating the last term:

$$A_{k+1}(n) = \sum_{j=0}^{k-1} a_j(n) + a_k(n) = A_k(n) + a_k(n).$$

Since $a_k(n) \geq 1$, it follows that

$$A_{k+1}(n) > A_k(n).$$

□

Observation 2.8. Although $A_k(n)$ always grows, what is relevant is not only its absolute growth, but its relative growth with respect to the critical threshold

$$k \log_2(3).$$

This threshold represents the amount of binary dissipation necessary to compensate for the accumulated ternary growth.

2.3 Ruiz Castillo residual debt

After k accelerated steps, the dominant ternary growth is given by 3^k. In binary scale,

$$3^k = 2^{k \log_2(3)}.$$

For this reason, the number

$$k \log_2(3)$$

represents the critical threshold of binary dissipation required to neutralize the dominant growth.

Definition 2.9 (Ruiz Castillo residual debt)**.** Let $n \in 2\mathbb{N} + 1$. The accumulated residual debt of order k is defined by

$$L_k(n) = k \log_2(3) - A_k(n).$$

The quantity $L_k(n)$ measures, on a binary logarithmic scale, the difference between the expansive demand of ternary growth and the binary dissipation effectively produced by the orbit.

Proposition 2.10 (Logarithmic form of the residual debt)**.** *For every $n \in 2\mathbb{N}+1$ and every $k \geq 1$, one has*

$$L_k(n) = \log_2\left(\frac{3^k}{2^{A_k(n)}}\right).$$

Proof. By properties of logarithms,

$$\log_2\left(\frac{3^k}{2^{A_k(n)}}\right) = \log_2(3^k) - \log_2(2^{A_k(n)}).$$

Since

$$\log_2(3^k) = k \log_2(3)$$

and

$$\log_2(2^{A_k(n)}) = A_k(n),$$

one obtains

$$\log_2\left(\frac{3^k}{2^{A_k(n)}}\right) = k \log_2(3) - A_k(n) = L_k(n).$$

□

Interpretation of the residual debt

The residual debt compares two accumulated magnitudes:

$$3^k \qquad \text{and} \qquad 2^{A_k(n)}.$$

If $L_k(n) > 0$, the orbit retains a dissipative deficit. If $L_k(n) < 0$, the orbit presents dissipative surplus. The residual debt therefore measures the relative pressure between ternary expansion and binary dissipation.

2.4 Sign of the residual debt

Proposition 2.11 (Interpretation of the sign of the residual debt)**.** *For every $n \in 2\mathbb{N}+1$ and every $k \geq 1$, one has:*

$$L_k(n) < 0 \quad \Longleftrightarrow \quad \frac{3^k}{2^{A_k(n)}} < 1,$$

and

$$L_k(n) > 0 \quad \Longleftrightarrow \quad \frac{3^k}{2^{A_k(n)}} > 1.$$

Proof. By the logarithmic form,

$$L_k(n) = \log_2\left(\frac{3^k}{2^{A_k(n)}}\right).$$

Since the function $\log_2(x)$ is strictly increasing and satisfies

$$\log_2(x) < 0 \Longleftrightarrow 0 < x < 1,$$

one has

$$L_k(n) < 0 \quad \Longleftrightarrow \quad \frac{3^k}{2^{A_k(n)}} < 1.$$

Similarly,

$$\log_2(x) > 0 \Longleftrightarrow x > 1,$$

so that

$$L_k(n) > 0 \quad \Longleftrightarrow \quad \frac{3^k}{2^{A_k(n)}} > 1.$$

$\square$

Corollary 2.12 (Dominant contraction)**.** *If*

$$L_k(n) < 0,$$

then

$$\frac{3^k}{2^{A_k(n)}} < 1.$$

Therefore, the dominant multiplicative component of the accelerated iteration is contractive.

Observation 2.13. Dominant contraction does not automatically imply effective descent of the orbit. The reason is that the complete form of $U^k(n)$ also contains an accumulated residual term associated with the $+1$ terms. This term will later be studied through the normalized residue $R_k(n)$.

2.5 Nonexistence of exact equilibrium

The ideal boundary between dissipative deficit and surplus would be given by

$$L_k(n) = 0.$$

However, this equality cannot occur for $k \geq 1$.

Proposition 2.14 (Nonexistence of exact equilibrium)**.** *For every $n \in 2\mathbb{N}+1$ and every $k \geq 1$, it is not the case that*

$$L_k(n) = 0.$$

Proof. Suppose, for contradiction, that there exist $n \in 2\mathbb{N}+1$ and $k \geq 1$ such that

$$L_k(n) = 0.$$

Then

$$k \log_2(3) - A_k(n) = 0,$$

whence

$$A_k(n) = k \log_2(3).$$

Raising 2 to both sides:

$$2^{A_k(n)} = 2^{k \log_2(3)} = 3^k.$$

But $A_k(n)$ is a positive integer, so $2^{A_k(n)}$ is a power of 2, whereas 3^k is a power of 3. By uniqueness of prime factorization, this is impossible for $k \geq 1$. □

Observation 2.15. Although exact equilibrium does not exist, nearly critical blocks may exist for which

$$|L_k(n)|$$

is small. These blocks represent zones of narrow competition between ternary growth and binary dissipation.

2.6 Residual classification of blocks

Definition 2.16 (Residual classification)**.** Let $n \in 2\mathbb{N}+1$ and $k \geq 1$. The initial block of length k is said to be:

(i) *residual expansive* if

$$L_k(n) > 0;$$

(ii) *residual dissipative* if

$$L_k(n) < 0;$$

(iii) *nearly critical* if, for some $\varepsilon > 0$,

$$|L_k(n)| < \varepsilon.$$

Synthesis

The Ruiz Castillo residual debt makes it possible to transform the analysis of the accelerated dynamics into a balance problem:

$$\text{accumulated ternary expansion} \quad \longleftrightarrow \quad \text{accumulated binary dissipation.}$$

This balance constitutes the basis for later introducing the normalized residue and the residual decomposition.

3 Exact affine representation

The residual debt $L_k(n)$ makes it possible to study the dominant multiplicative factor of the accelerated dynamics. Indeed,

$$2^{L_k(n)} = \frac{3^k}{2^{A_k(n)}}.$$

However, this quotient does not completely describe the iteration $U^k(n)$, because each application of the accelerated map comes from an affine transformation of the form

$$n \longmapsto 3n + 1.$$

The additive term $+1$, although locally small in comparison with $3n$, propagates under successive iterations and generates a nontrivial arithmetic accumulation.

For this reason, the accelerated dynamics cannot be reduced only to the balance between 3^k and $2^{A_k(n)}$. An accumulated residual term must be incorporated to record the memory of all affine increments produced during the first k steps.

Motivation

The residual debt controls the dominant quotient

$$\frac{3^k}{2^{A_k(n)}}.$$

But the complete iteration also contains the accumulation of the $+1$ terms. This accumulation produces an arithmetic residue that will be denoted by $B_k(n)$.

3.1 Construction of the accumulated residual term

Let $n \in 2\mathbb{N} + 1$. Recall that the accelerated orbit satisfies

$$n_{j+1} = \frac{3n_j + 1}{2^{a_j(n)}},$$

where

$$n_j = U^j(n), \qquad a_j(n) = \nu_2(3U^j(n) + 1).$$

After one step one has

$$U(n) = \frac{3n + 1}{2^{a_0(n)}}.$$

After two steps,

$$U^2(n) = \frac{3U(n) + 1}{2^{a_1(n)}} = \frac{3\left(\frac{3n+1}{2^{a_0(n)}}\right) + 1}{2^{a_1(n)}}.$$

By unifying denominators:

$$U^2(n) = \frac{3^2 n + 3 + 2^{a_0(n)}}{2^{a_0(n)+a_1(n)}}.$$

This calculation shows that the numerator contains the dominant term $3^2 n$, but also a residual part:

$$3 + 2^{a_0(n)}.$$

Analogously, at each step new residual terms are incorporated, weighted by powers of 3 and accumulated powers of 2.

This observation motivates the following representation.

3.2 Exact affine representation

Proposition 3.1 (Exact affine representation)**.** *For every $n \in 2\mathbb{N}+1$ and every $k \geq 1$, there exists a positive integer $B_k(n)$ such that*

$$U^k(n) = \frac{3^k n + B_k(n)}{2^{A_k(n)}}.$$

Moreover,

$$B_1(n) = 1,$$

and

$$B_{k+1}(n) = 3B_k(n) + 2^{A_k(n)}.$$

Proof. We proceed by induction on k.

For $k = 1$, by definition of the accelerated map:

$$U(n) = \frac{3n+1}{2^{a_0(n)}}.$$

Since

$$A_1(n) = a_0(n),$$

one obtains

$$U(n) = \frac{3n+1}{2^{A_1(n)}}.$$

Therefore, the formula holds by taking

$$B_1(n) = 1.$$

Now suppose that for some $k \geq 1$,

$$U^k(n) = \frac{3^k n + B_k(n)}{2^{A_k(n)}}.$$

Applying the accelerated map again:

$$U^{k+1}(n) = U(U^k(n)) = \frac{3U^k(n) + 1}{2^{a_k(n)}}.$$

Substituting the inductive hypothesis:

$$U^{k+1}(n) = \frac{3\left(\frac{3^k n + B_k(n)}{2^{A_k(n)}}\right) + 1}{2^{a_k(n)}}.$$

Writing the numerator with a common denominator:

$$3\left(\frac{3^k n + B_k(n)}{2^{A_k(n)}}\right) + 1 = \frac{3^{k+1} n + 3B_k(n) + 2^{A_k(n)}}{2^{A_k(n)}}.$$

Therefore,

$$U^{k+1}(n) = \frac{3^{k+1} n + 3B_k(n) + 2^{A_k(n)}}{2^{A_k(n)} 2^{a_k(n)}}.$$

Since

$$A_{k+1}(n) = A_k(n) + a_k(n),$$

one obtains

$$U^{k+1}(n) = \frac{3^{k+1} n + B_{k+1}(n)}{2^{A_{k+1}(n)}},$$

where

$$B_{k+1}(n) = 3B_k(n) + 2^{A_k(n)}.$$

This completes the induction. □

Observation 3.2. The exact affine representation separates the accelerated iteration into two parts:

$$3^k n$$

corresponds to pure multiplicative growth, while

$$B_k(n)$$

collects the accumulation of all additive terms generated by the transformations $3x + 1$. For this reason, $B_k(n)$ can be interpreted as a residual memory of the orbit.

3.3 Explicit formula for the residual term

The recurrence

$$B_{k+1}(n) = 3B_k(n) + 2^{A_k(n)}$$

allows one to obtain a closed expression for $B_k(n)$. This expression clearly shows the way in which residual memory is organized in the accelerated orbit.

Proposition 3.3 (Explicit formula for the residual term)**.** *For every $k \geq 1$, one has*

$$B_k(n) = \sum_{i=0}^{k-1} 3^{k-1-i} 2^{A_i(n)}.$$

Proof. We proceed by induction.

For $k = 1$, using the convention $A_0(n) = 0$, one has

$$\sum_{i=0}^{0} 3^{0-i} 2^{A_i(n)} = 3^0 2^{A_0(n)} = 1.$$

Since $B_1(n) = 1$, the formula is valid for $k = 1$.

Now suppose that for some $k \geq 1$,

$$B_k(n) = \sum_{i=0}^{k-1} 3^{k-1-i} 2^{A_i(n)}.$$

Then, using the recurrence:

$$B_{k+1}(n) = 3B_k(n) + 2^{A_k(n)}.$$

Substituting the inductive hypothesis:

$$B_{k+1}(n) = 3 \sum_{i=0}^{k-1} 3^{k-1-i} 2^{A_i(n)} + 2^{A_k(n)}.$$

Distributing the factor 3:

$$B_{k+1}(n) = \sum_{i=0}^{k-1} 3^{k-i} 2^{A_i(n)} + 2^{A_k(n)}.$$

Since

$$2^{A_k(n)} = 3^0 2^{A_k(n)},$$

one obtains

$$B_{k+1}(n) = \sum_{i=0}^{k} 3^{k-i} 2^{A_i(n)}.$$

By induction, the formula is proved. □

Interpretation of the residual term

The formula
$$B_k(n) = \sum_{i=0}^{k-1} 3^{k-1-i} 2^{A_i(n)}$$
shows that each residual term generated at an earlier step is later amplified by a power of 3. Thus, $B_k(n)$ is not an accidental residue, but an accumulated and weighted memory of the affine dynamics.

3.4 First properties of $B_k(n)$

Proposition 3.4 (Positivity and integrality)**.** *For every $n \in 2\mathbb{N}+1$ and every $k \geq 1$,*
$$B_k(n) \in \mathbb{N}$$
and
$$B_k(n) > 0.$$

Proof. The explicit formula
$$B_k(n) = \sum_{i=0}^{k-1} 3^{k-1-i} 2^{A_i(n)}$$
is a finite sum of positive integers. Therefore, $B_k(n)$ is a positive integer. □

Proposition 3.5 (Symbolic dependence of the residue)**.** *The term $B_k(n)$ depends on the finite valuation word*
$$(a_0(n), a_1(n), \ldots, a_{k-1}(n))$$
only through the partial sums
$$A_i(n) = \sum_{j=0}^{i-1} a_j(n).$$

Proof. The explicit formula for $B_k(n)$ is
$$B_k(n) = \sum_{i=0}^{k-1} 3^{k-1-i} 2^{A_i(n)}.$$

Each $A_i(n)$ is determined by the finite word
$$(a_0(n), a_1(n), \ldots, a_{i-1}(n)).$$

Therefore, $B_k(n)$ is determined by the finite valuation word up to time $k-1$. $\square$

Observation 3.6. Although $B_k(n)$ appears within the expression for $U^k(n)$, its internal structure is governed by the valuation word. This connects the affine representation with the symbolic dynamics induced by the 2-adic valuations.

3.5 Illustrative example

Consider $k = 3$. The explicit formula gives

$$B_3(n) = 3^2 2^{A_0(n)} + 3^1 2^{A_1(n)} + 3^0 2^{A_2(n)}.$$

Since

$$A_0(n) = 0, \qquad A_1(n) = a_0(n), \qquad A_2(n) = a_0(n) + a_1(n),$$

one obtains

$$B_3(n) = 9 + 3 \cdot 2^{a_0(n)} + 2^{a_0(n)+a_1(n)}.$$

Thus,

$$U^3(n) = \frac{27n + 9 + 3 \cdot 2^{a_0(n)} + 2^{a_0(n)+a_1(n)}}{2^{a_0(n)+a_1(n)+a_2(n)}}.$$

This example shows that the residual terms are not added homogeneously. Each contribution is weighted by the temporal position in which it was produced and by the accumulated dissipation up to that moment.

3.6 Importance of the affine representation

The exact affine representation makes it possible to understand why the study of residual debt is not sufficient to describe the accelerated dynamics completely. Indeed, the formula

$$U^k(n) = \frac{3^k n + B_k(n)}{2^{A_k(n)}}$$

can be rewritten as

$$U^k(n) = \frac{3^k}{2^{A_k(n)}} n + \frac{B_k(n)}{2^{A_k(n)}}.$$

The first term is governed by the residual debt:

$$\frac{3^k}{2^{A_k(n)}} = 2^{L_k(n)}.$$

The second term, in contrast, requires introducing a new quantity:

$$R_k(n) = \frac{B_k(n)}{2^{A_k(n)}}.$$

Step toward the residual decomposition

The exact affine representation naturally leads to the identity

$$U^k(n) = 2^{L_k(n)} n + R_k(n).$$

This identity separates the dominant multiplicative component from the normalized residual memory, and constitutes the basis of the Ruiz Castillo residual decomposition.

4 Ruiz Castillo residual decomposition

The exact affine representation obtained in the previous section reveals that the accelerated Collatz dynamics simultaneously possesses a multiplicative component and an additive component. However, both appear mixed within the expression

$$U^k(n) = \frac{3^k n + B_k(n)}{2^{A_k(n)}}.$$

From a dynamical perspective, it is convenient to explicitly separate both contributions. This separation leads to a new formulation of the problem, in which the evolution of the orbit is decomposed into a dominant force associated with the balance between ternary growth and binary dissipation, and a residual force associated with the accumulated memory of the affine terms.

The introduction of this separation constitutes one of the central elements of the present research.

4.1 Normalized residue

Definition 4.1 (Ruiz Castillo normalized residue)**.** Let $n \in 2\mathbb{N}+1$ and $k \geq 1$. The normalized residue is defined by

$$R_k(n) = \frac{B_k(n)}{2^{A_k(n)}}.$$

Observation 4.2. The term

$$B_k(n)$$

represents the accumulated memory of all additive terms generated by the transformations $3x + 1$. However, the effective size of this memory does not depend only on $B_k(n)$, but also on the accumulated binary dissipation.

Normalization by

$$2^{A_k(n)}$$

makes it possible to measure the residual contribution on the same dynamical scale that governs the accelerated orbit.

4.2 Fundamental decomposition theorem

Theorem 4.3 (Ruiz Castillo Residual Decomposition Theorem)**.** *For every* $n \in 2\mathbb{N} + 1$ *and every* $k \geq 1$*,*

$$U^k(n) = 2^{L_k(n)}\, n + R_k(n),$$

where

$$L_k(n) = k \log_2(3) - A_k(n),$$

and

$$R_k(n) = \frac{B_k(n)}{2^{A_k(n)}}.$$

Proof. We start from the exact affine representation

$$U^k(n) = \frac{3^k n + B_k(n)}{2^{A_k(n)}}.$$

Separating terms,

$$U^k(n) = \frac{3^k}{2^{A_k(n)}}\, n + \frac{B_k(n)}{2^{A_k(n)}}.$$

By definition of the residual debt,

$$L_k(n) = k \log_2(3) - A_k(n).$$

Therefore,

$$2^{L_k(n)} = 2^{k \log_2(3) - A_k(n)} = \frac{3^k}{2^{A_k(n)}}.$$

Likewise,

$$R_k(n) = \frac{B_k(n)}{2^{A_k(n)}}.$$

Substituting both expressions,

$$U^k(n) = 2^{L_k(n)}n + R_k(n).$$

□

4.3 Dynamical interpretation

Structural reading

The Ruiz Castillo residual decomposition expresses the accelerated orbit as the sum of two mathematically distinct mechanisms:

$$U^k(n) = \underbrace{2^{L_k(n)}n}_{\text{dominant component}} + \underbrace{R_k(n)}_{\text{residual component}}.$$

The dominant component is governed exclusively by the balance between:

$$3^k \qquad \text{and} \qquad 2^{A_k(n)}.$$

The residual component contains all the accumulated memory of the affine terms $+1$.

From this perspective, the Collatz problem ceases to be seen only as a succession of arithmetic operations and comes to be interpreted as an interaction between two forces:

expansion–contraction

and

residual memory.

The first depends on the residual debt $L_k(n)$; the second depends on the normalized residue $R_k(n)$.

4.4 Descent criterion

The previous decomposition allows an exact criterion for effective descent to be formulated.

Proposition 4.4 (Residual descent criterion)**.** *For every $n \in 2\mathbb{N}+1$,*

$$U^k(n) < n$$

if and only if

$$2^{L_k(n)} + \frac{R_k(n)}{n} < 1.$$

Proof. Starting from

$$U^k(n) = 2^{L_k(n)} n + R_k(n),$$

the condition

$$U^k(n) < n$$

is equivalent to

$$2^{L_k(n)} n + R_k(n) < n.$$

Dividing by $n > 0$,

$$2^{L_k(n)} + \frac{R_k(n)}{n} < 1.$$

□

Observation 4.5. This identity shows that negative residual debt does not by itself guarantee descent of the orbit.

Even when

$$L_k(n) < 0,$$

the residual term

$$R_k(n)$$

can partially compensate for the dominant contraction.

For this reason, the simultaneous study of

$$L_k(n) \quad \text{and} \quad R_k(n)$$

is indispensable.

4.5 Principle of residual compensation

The decomposition obtained naturally leads to the following heuristic principle.

Principle of residual compensation

The Collatz Conjecture can be reinterpreted as the assertion that accumulated binary dissipation simultaneously dominates:

(i) multiplicative growth

measured by the residual debt $L_k(n)$,
and

(ii) accumulated affine memory

measured by the normalized residue $R_k(n)$.
Consequently, an orbit descends when the contraction induced by $L_k(n)$ effectively exceeds the residual contribution $R_k(n)$.

4.6 Originality of the formulation

Observation 4.6 (Ruiz Castillo conceptual contribution). The joint introduction of the objects

$$L_k(n), \qquad D_k(n) = \frac{L_k(n)}{k}, \qquad R_k(n),$$

together with the identity

$$U^k(n) = 2^{L_k(n)} n + R_k(n),$$

provides a new conceptual architecture for the study of accelerated dynamics.

Whereas the classical literature focuses mainly on accumulated 2-adic valuations or on statistical properties of the orbits, this formulation explicitly separates:

1. the dominant force of expansion or contraction;
2. the residual memory generated by the affine terms;
3. the interaction between both contributions.

This decomposition makes it possible to transfer part of the problem toward the joint study of residual drift and the asymptotic behavior of the normalized residue.

5 Symbolic formula for the normalized residue

The Ruiz Castillo residual decomposition introduces the normalized residue

$$R_k(n) = \frac{B_k(n)}{2^{A_k(n)}}.$$

This quantity measures the effective contribution of the accumulated residual term after it has been subjected to the total binary dissipation of the block. To understand its internal structure, it is not enough to observe the total sum $A_k(n)$; it is necessary to study how the valuations are distributed within the finite word

$$(a_0(n), a_1(n), \ldots, a_{k-1}(n)).$$

In this section it is shown that $R_k(n)$ can be expressed as a symbolic functional of this word. This formulation is relevant because it allows the analysis of the residue to be transferred from the numerical orbit to the combinatorial and dissipative structure of the 2-adic valuations.

5.1 Symbolic form of the residue

Proposition 5.1 (Symbolic formula for $R_k(n)$)**.** *For every $n \in 2\mathbb{N}+1$ and every $k \geq 1$, one has*

$$R_k(n) = \sum_{i=0}^{k-1} 3^{k-1-i} 2^{A_i(n) - A_k(n)}.$$

Equivalently,

$$R_k(n) = \sum_{i=0}^{k-1} 3^{k-1-i} 2^{-\sum_{j=i}^{k-1} a_j(n)}.$$

Proof. By definition of the normalized residue,

$$R_k(n) = \frac{B_k(n)}{2^{A_k(n)}}.$$

Moreover, by the explicit formula for the accumulated residual term,

$$B_k(n) = \sum_{i=0}^{k-1} 3^{k-1-i} 2^{A_i(n)}.$$

Substituting this expression in the definition of $R_k(n)$, one obtains

$$R_k(n) = \frac{1}{2^{A_k(n)}} \sum_{i=0}^{k-1} 3^{k-1-i} 2^{A_i(n)}.$$

Distributing the factor $2^{-A_k(n)}$ in the sum:

$$R_k(n) = \sum_{i=0}^{k-1} 3^{k-1-i} 2^{A_i(n)-A_k(n)}.$$

Now, by definition of $A_k(n)$,

$$A_k(n) = \sum_{j=0}^{k-1} a_j(n),$$

whereas

$$A_i(n) = \sum_{j=0}^{i-1} a_j(n).$$

Therefore,

$$A_k(n) - A_i(n) = \sum_{j=i}^{k-1} a_j(n).$$

It follows that

$$A_i(n) - A_k(n) = -\sum_{j=i}^{k-1} a_j(n).$$

Substituting in the previous expression:

$$R_k(n) = \sum_{i=0}^{k-1} 3^{k-1-i} 2^{-\sum_{j=i}^{k-1} a_j(n)}.$$

□

5.2 Interpretation of the symbolic formula

The expression

$$R_k(n) = \sum_{i=0}^{k-1} 3^{k-1-i} 2^{-\sum_{j=i}^{k-1} a_j(n)}$$

shows that each residual contribution depends on two opposing factors.

On the one hand, the factor

$$3^{k-1-i}$$

indicates how much the residue generated at time i is amplified by the subsequent ternary multiplications. The earlier a residual contribution appears, the greater the power of 3 that amplifies it.

On the other hand, the factor

$$2^{-\sum_{j=i}^{k-1} a_j(n)}$$

measures the binary dissipation acting from time i to time $k-1$. The greater the accumulated dissipation in that tail, the smaller the effective contribution of that residue.

Dynamical reading

Each term of $R_k(n)$ has the form

$$\text{subsequent ternary amplification} \quad \times \quad \text{tail binary dissipation.}$$

Therefore, the normalized residue does not depend only on how much total dissipation exists, but on how that dissipation is temporally distributed within the block.

This observation is essential. Two blocks may have the same accumulated dissipation $A_k(n)$, but produce different values of $R_k(n)$, because the normalized residue is sensitive to the order of the valuations. It is not the same to concentrate dissipation at the beginning of the block as to concentrate it at the end.

5.3 Dependence on dissipative tails

The symbolic formula shows that the normalized residue is governed by tail sums:

$$\sum_{j=i}^{k-1} a_j(n).$$

These sums measure the dissipation available after a residual contribution has been generated.

Definition 5.2 (Tail dissipation)**.** For $0 \leq i < k$, the tail dissipation from i to $k-1$ is defined by

$$A_{i,k}(n) = \sum_{j=i}^{k-1} a_j(n).$$

With this notation,

$$R_k(n) = \sum_{i=0}^{k-1} 3^{k-1-i} 2^{-A_{i,k}(n)}.$$

Observation 5.3. The normalized residue is determined by the set of tail dissipations

$$A_{0,k}(n), A_{1,k}(n), \ldots, A_{k-1,k}(n).$$

These quantities contain more information than the total dissipation $A_k(n)$, because they record the temporal distribution of the dissipation.

5.4 Symbolic functional of the valuation word

Proposition 5.4 (Symbolic dependence)**.** *The normalized residue $R_k(n)$ depends only on the finite valuation word*

$$(a_0(n), a_1(n), \dots, a_{k-1}(n)).$$

In particular, if two accelerated orbits have the same valuation word of length k, then they have the same value of R_k.

Proof. The formula

$$R_k(n) = \sum_{i=0}^{k-1} 3^{k-1-i} 2^{-\sum_{j=i}^{k-1} a_j(n)}$$

expresses $R_k(n)$ solely in terms of the values

$$a_0(n), a_1(n), \dots, a_{k-1}(n).$$

Therefore, if two orbits have the same valuation word of length k, all tail sums coincide and, consequently, the value of R_k also coincides. □

Symbolic consequence

The normalized residue $R_k(n)$ should not be understood only as an algebraic term. It is a symbolic functional of the dissipative structure of the orbit:

$$R_k = \mathcal{R}(a_0, a_1, \dots, a_{k-1}).$$

This observation connects the residual decomposition with the symbolic dynamics induced by the 2-adic valuations.

5.5 Sensitivity to the order of the valuations

The dependence of $R_k(n)$ on dissipative tails implies that the order of the valuations is relevant. The normalized residue is not determined only by the multiset

$$\{a_0(n), a_1(n), \dots, a_{k-1}(n)\},$$

but by the ordered word.

Proposition 5.5 (Sensitivity to order)**.** *There exist valuation words with the same total sum A_k, but with distinct normalized residues.*

Proof. Consider $k = 2$. For a word (a_0, a_1), one has

$$R_2 = 3^1 2^{-(a_0+a_1)} + 3^0 2^{-a_1} = 3 \cdot 2^{-(a_0+a_1)} + 2^{-a_1}.$$

Compare the words

$$(1,3) \qquad \text{and} \qquad (3,1).$$

Both have total sum

$$A_2 = 4.$$

For $(1,3)$,

$$R_2 = 3 \cdot 2^{-4} + 2^{-3} = \frac{3}{16} + \frac{1}{8} = \frac{5}{16}.$$

For $(3,1)$,

$$R_2 = 3 \cdot 2^{-4} + 2^{-1} = \frac{3}{16} + \frac{1}{2} = \frac{11}{16}.$$

Therefore, although the total dissipation coincides, the normalized residue is different. $\square$

Observation 5.6. This result shows that the residual debt $L_k(n)$, by depending only on $A_k(n)$, does not capture the entire dynamical structure of the block. The normalized residue $R_k(n)$, in contrast, distinguishes the temporal distribution of the valuations.

5.6 Conclusion of the section

The symbolic formula for $R_k(n)$ reveals that accumulated residual memory is governed by the complete structure of the valuation word. Whereas residual debt measures the global balance between growth and dissipation, the normalized residue measures how dissipation is distributed throughout the block and how that distribution affects residual contributions.

Synthesis

The residual debt $L_k(n)$ measures the global balance:

$$3^k \quad \text{versus} \quad 2^{A_k(n)}.$$

The normalized residue $R_k(n)$ measures structured memory:

$$\sum_{i=0}^{k-1} 3^{k-1-i} 2^{-A_{i,k}(n)}.$$

Therefore, the study of effective descent requires simultaneously analyzing the global average of the valuations and their temporal distribution.

6 Dynamical structure of the residual decomposition

The identity

$$U^k(n) = 2^{L_k(n)} n + R_k(n)$$

allows three scenarios to be distinguished.

(i) Dominant expansive regime:

$$L_k(n) > 0.$$

In this case, $2^{L_k(n)} > 1$, so the dominant component amplifies the initial value.

(ii) Dominant contractive regime:

$$L_k(n) < 0.$$

In this case, $2^{L_k(n)} < 1$, so the dominant component contracts the initial value.

(iii) Nearly critical regime:

$$|L_k(n)| \approx 0.$$

In this case, the dominant factor approaches 1, and the behavior of the orbit depends more strongly on the normalized residue.

Observation 6.1. The nearly critical regime is especially delicate, since residual debt close to zero indicates approximate equilibrium between ternary growth and binary dissipation. In that zone, small variations in $R_k(n)$ can determine whether effective descent exists or not.

Figure 1 conceptually illustrates the separation between the dominant component and the normalized residue. The graph does not represent real computational data; its function is to show the algebraic structure of the decomposition.

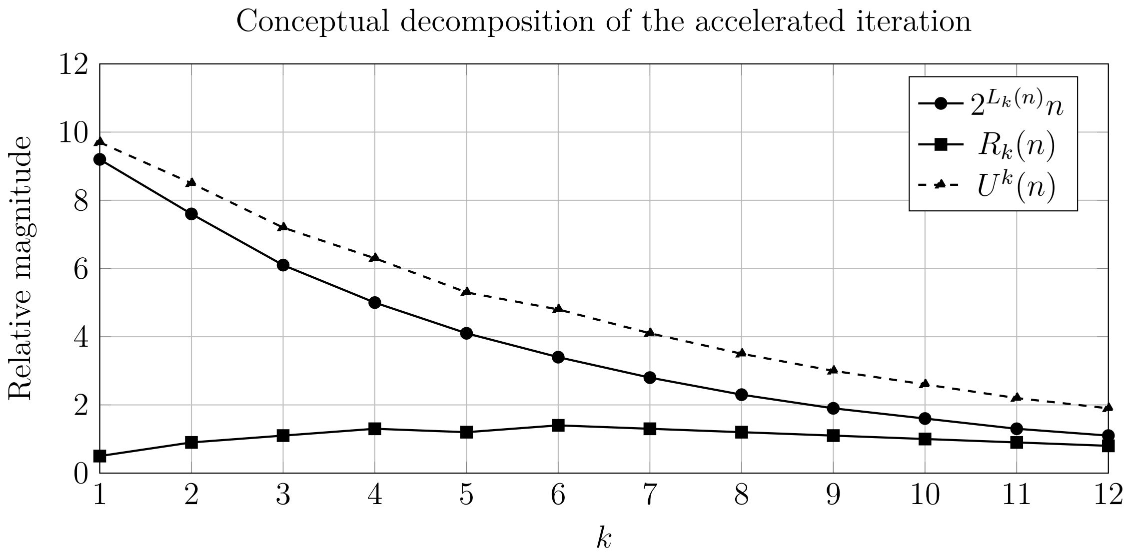

Figure 1: Conceptual visualization of the Ruiz Castillo residual decomposition.

The mathematical reading of Figure 1 is the following: even when the dominant component decreases, the normalized residue can delay effective descent. For this reason, the study of $R_k(n)$ is not secondary, but central for connecting dissipative pressure with the descent of orbits.

7 Dissipative bounds for $R_k(n)$

7.1 Basic universal bound

Proposition 7.1 (Basic universal bound)**.** *For every* $n \in 2\mathbb{N}+1$ *and every* $k \geq 1$,

$$R_k(n) \leq \left(\frac{3}{2}\right)^k - 1.$$

Proof. From the symbolic formula,

$$R_k(n) = \sum_{i=0}^{k-1} 3^{k-1-i} 2^{-\sum_{j=i}^{k-1} a_j(n)}.$$

Since $a_j(n) \geq 1$, one has

$$\sum_{j=i}^{k-1} a_j(n) \geq k - i.$$

Thus,

$$R_k(n) \leq \sum_{i=0}^{k-1} 3^{k-1-i} 2^{-(k-i)}.$$

With the change $r = k - i$:

$$R_k(n) \leq \sum_{r=1}^{k} 3^{r-1} 2^{-r} = \frac{1}{2} \sum_{r=1}^{k} \left(\frac{3}{2}\right)^{r-1}.$$

By the formula for the geometric sum,

$$R_k(n) \leq \left(\frac{3}{2}\right)^{k} - 1.$$

□

7.2 Bound under average tail dissipation

Definition 7.2 (Average tail dissipation)**.** For $0 \leq i < k$, define

$$\overline{a}_{i,k}(n) = \frac{1}{k-i} \sum_{j=i}^{k-1} a_j(n).$$

Definition 7.3 (Uniformly dissipative block)**.** A block of length k is said to be uniformly dissipative with parameter α if

$$\overline{a}_{i,k}(n) \geq \alpha$$

for every

$$0 \leq i < k.$$

Proposition 7.4 (Bound by average tail dissipation)**.** *Suppose that there exists $\alpha > 0$ such that for every $0 \leq i < k$,*

$$\overline{a}_{i,k}(n) \geq \alpha.$$

Then

$$R_k(n) \leq \sum_{r=1}^{k} 3^{r-1} 2^{-\alpha r}.$$

Proof. The hypothesis implies

$$\sum_{j=i}^{k-1} a_j(n) \geq \alpha(k - i).$$

Therefore,

$$2^{-\sum_{j=i}^{k-1} a_j(n)} \leq 2^{-\alpha(k-i)}.$$

Substituting in the formula for $R_k(n)$:

$$R_k(n) \leq \sum_{i=0}^{k-1} 3^{k-1-i} 2^{-\alpha(k-i)}.$$

With $r = k - i$, one obtains

$$R_k(n) \leq \sum_{r=1}^{k} 3^{r-1} 2^{-\alpha r}.$$

□

Corollary 7.5 (Uniform boundedness under strong dissipation)**.** *If*

$$\alpha > \log_2(3),$$

then

$$R_k(n) \leq \frac{2^{-\alpha}}{1 - \frac{3}{2^\alpha}}.$$

Proof. From the previous proposition:

$$R_k(n) \leq \sum_{r=1}^{k} 3^{r-1} 2^{-\alpha r}.$$

Each term can be written as

$$3^{r-1} 2^{-\alpha r} = 2^{-\alpha} \left(\frac{3}{2^\alpha} \right)^{r-1}.$$

If $\alpha > \log_2(3)$, then

$$\frac{3}{2^\alpha} < 1.$$

Therefore,

$$R_k(n) \leq 2^{-\alpha} \sum_{r=1}^{\infty} \left(\frac{3}{2^\alpha} \right)^{r-1} = \frac{2^{-\alpha}}{1 - \frac{3}{2^\alpha}}.$$

□

Key result

If all final tails of the valuation word have dissipative average greater than

$$\log_2(3),$$

then the normalized residue $R_k(n)$ remains uniformly bounded.

Figure 2 shows the behavior of the bound

$$C(\alpha) = \frac{2^{-\alpha}}{1 - \frac{3}{2^\alpha}},$$

valid for $\alpha > \log_2(3)$.

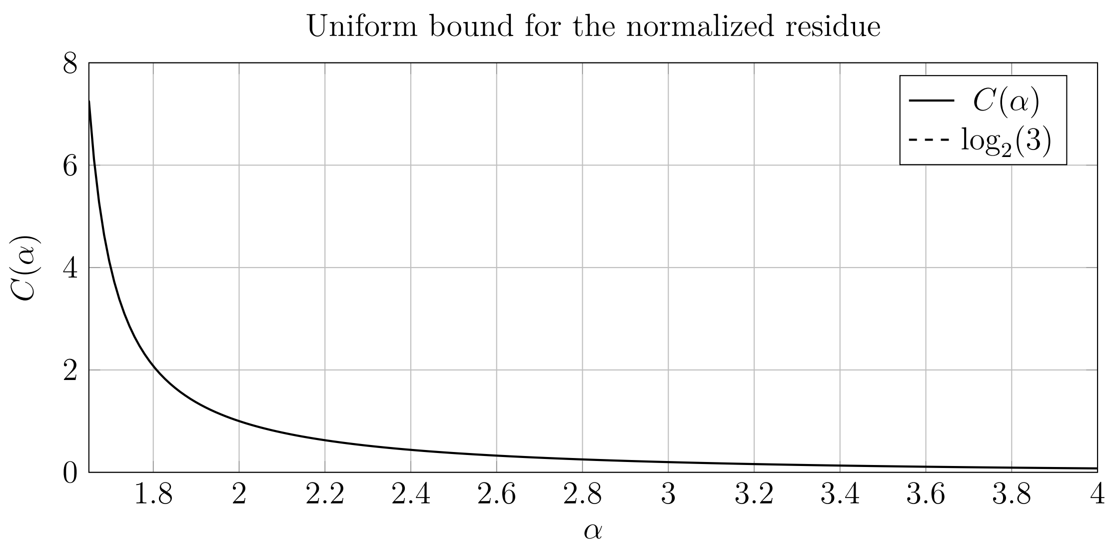

Figure 2: Conceptual behavior of the uniform bound for the normalized residue under average tail dissipation.

Figure 2 shows that the bound grows without limit when α approaches $\log_2(3)$ from the right. This indicates that nearly critical blocks do not offer sufficient control over the normalized residue. In contrast, when α increases, the residue becomes progressively more controlled.

8 Residual regimes and criterion for effective descent

The Ruiz Castillo residual decomposition

$$U^k(n) = 2^{L_k(n)} n + R_k(n)$$

allows the problem of descent of an accelerated orbit to be reinterpreted as a problem of equilibrium between two magnitudes:

$$\underbrace{2^{L_k(n)} n}_{\text{dominant component}} \qquad \text{and} \qquad \underbrace{R_k(n)}_{\text{residual component}} .$$

The first determines the principal tendency of expansion or contraction induced by the balance between ternary growth and binary dissipation. The second represents the accumulated memory of the affine terms.

Effective descent occurs only when the sum of both components remains below the initial value n.

8.1 Available residual descent margin

Definition 8.1 (Available residual margin)**.** Let $n \in 2\mathbb{N} + 1$ and $k \geq 1$.

The available residual margin is defined by

$$M_k(n) = \left(1 - 2^{L_k(n)}\right) n.$$

Observation 8.2. The quantity $M_k(n)$ measures the maximum capacity that the normalized residue has in order not to destroy the contraction induced by the dominant component.

Geometrically,

$$M_k(n) = n - 2^{L_k(n)} n$$

represents the distance between the initial value n and the effective trajectory

$$2^{L_k(n)} n.$$

8.2 Residual regimes

Definition 8.3 (Descending residual regime)**.** An accelerated block of length k is in a descending residual regime when

$$R_k(n) < M_k(n).$$

Equivalently,

$$R_k(n) < \left(1 - 2^{L_k(n)}\right) n.$$

Definition 8.4 (Critical residual regime)**.** An accelerated block of length k is in a critical residual regime when

$$R_k(n) = M_k(n).$$

Equivalently,

$$R_k(n) = \left(1 - 2^{L_k(n)}\right) n.$$

Definition 8.5 (Non-descending residual regime)**.** An accelerated block of length k is in a non-descending residual regime when

$$R_k(n) > M_k(n).$$

Equivalently,

$$R_k(n) > \left(1 - 2^{L_k(n)}\right) n.$$

Observation 8.6. The previous definition completely classifies the local behavior of an accelerated block.

The descending regime produces effective descent.

The critical regime corresponds to a situation of exact equilibrium.

The non-descending regime prevents descent in that block.

8.3 Exact descent criterion

Theorem 8.7 (Ruiz Castillo residual criterion for effective descent)**.** *Let* $n \in 2\mathbb{N} + 1$ *and* $k \geq 1$.

Then

$$U^k(n) < n$$

if and only if

$$R_k(n) < \left(1 - 2^{L_k(n)}\right) n.$$

Proof. By the residual decomposition,

$$U^k(n) = 2^{L_k(n)} n + R_k(n).$$

Suppose first that

$$R_k(n) < \left(1 - 2^{L_k(n)}\right) n.$$

Then

$$U^k(n) < 2^{L_k(n)} n + \left(1 - 2^{L_k(n)}\right) n.$$

After simplification,

$$U^k(n) < n.$$

Conversely, suppose that

$$U^k(n) < n.$$

Substituting the residual decomposition,

$$2^{L_k(n)} n + R_k(n) < n.$$

Subtracting $2^{L_k(n)} n$,

$$R_k(n) < \left(1 - 2^{L_k(n)}\right) n.$$

Therefore, both conditions are equivalent.

$\square$

8.4 Geometric interpretation

Geometric interpretation of the criterion

The residual debt $L_k(n)$ determines an effective trajectory

$$\Phi_k(n) = 2^{L_k(n)} n.$$

Effective descent depends on whether the residual memory $R_k(n)$ manages to remain within the available region

$$0 < R_k(n) < n - \Phi_k(n).$$

In other words, the dominant contraction generates a dissipative reserve

$$M_k(n) = n - \Phi_k(n),$$

and descent occurs exactly when the normalized residue does not completely consume that reserve.

8.5 Contractive case

Corollary 8.8. *If*

$$L_k(n) < 0$$

and

$$R_k(n) < \left(1 - 2^{L_k(n)}\right) n,$$

then

$$U^k(n) < n.$$

Observation 8.9. This result shows that negative residual debt is not sufficient to guarantee descent.

The truly relevant condition is the simultaneous control of

$$L_k(n)$$

and

$$R_k(n).$$

For this reason, the theory of residual drift must necessarily be complemented by a theory of the normalized residue.

8.6 Fundamental principle

Principle of residual descent

The Collatz Conjecture can be reinterpreted as the assertion that every accelerated orbit eventually generates blocks for which

$$R_k(n) < \left(1 - 2^{L_k(n)}\right) n.$$

Under this formulation, the problem ceases to depend exclusively on the sign of the residual debt and comes to depend on the equilibrium between dissipative pressure and residual memory.

9 Conclusion

The advance developed in this work introduces a new structural piece within the residual theory of accelerated Collatz dynamics: the *Ruiz Castillo residual decomposition*,

$$\boxed{U^k(n) = 2^{L_k(n)} n + R_k(n).}$$

This identity makes it possible to formally separate the accelerated iteration into two mathematically differentiated components. The first,

$$2^{L_k(n)} n,$$

corresponds to the dominant factor of expansion or contraction, determined by the residual debt

$$L_k(n) = k \log_2(3) - A_k(n).$$

The second,

$$R_k(n),$$

corresponds to the normalized residual memory, generated by the accumulation of the additive terms $+1$ present in each application of the odd transformation.

With this separation, the analysis of the accelerated dynamics ceases to depend only on the comparison between

$$3^k \qquad \text{and} \qquad 2^{A_k(n)}.$$

That comparison remains fundamental, because it determines the sign of the residual debt and the contractive or expansive character of the dominant factor. However, effective descent of an orbit requires a second level of analysis: quantitative control of the normalized residue.

Central contribution

The main contribution consists in showing that negative residual drift is not by itself sufficient to guarantee effective descent. Descent requires accumulated dissipative pressure to dominate simultaneously the ternary growth and the affine residual memory:

$$R_k(n) < \left(1 - 2^{L_k(n)}\right) n.$$

Consequently, the problem is reorganized into two complementary levels:

Level I: proving dissipative pressure or negative residual debt,

Level II: controlling the normalized residual memory.

The first level studies the behavior of $L_k(n)$, the residual drift, and the relationship between ternary growth and binary dissipation. The second level studies $R_k(n)$, its dependence on the valuation word, and its capacity to delay or prevent effective descent.

This reorganization provides a finer reading of the problem. Residual debt measures the global balance:

$$3^k \quad \text{versus} \quad 2^{A_k(n)}.$$

The normalized residue, in contrast, measures the internal structure of the accumulated memory:

$$R_k(n) = \sum_{i=0}^{k-1} 3^{k-1-i} 2^{-\sum_{j=i}^{k-1} a_j(n)}.$$

Therefore, whereas $L_k(n)$ depends on the total accumulated dissipation, $R_k(n)$ depends on the temporal distribution of the valuations within the block. This distinction is essential, because two blocks with the same total sum of valuations can produce distinct normalized residues.

Conceptual synthesis

The accelerated dynamics can be interpreted as a system with:

$$\text{dominant contraction} + \text{residual memory}.$$

The Collatz Conjecture, from this perspective, requires understanding how binary dissipation controls not only the multiplicative factor, but also the memory generated by the affine structure of the transformation $3n + 1$.

The present work does not claim to assert a final proof of the Collatz Conjecture. Its purpose is to construct a formal architecture that makes it possible to study the transition between dissipative compensation and effective descent. In this framework, the normalized residue $R_k(n)$ becomes a central mathematical object: it is not an accessory term, but the bridge between the dissipative pressure measured by the residual debt and the actual descent of the accelerated orbit.

The line of research opened by this decomposition consists in studying conditions on the valuation word

$$(a_0(n), a_1(n), \ldots, a_{k-1}(n))$$

that simultaneously guarantee:

$$L_k(n) < 0$$

and

$$R_k(n) < \left(1 - 2^{L_k(n)}\right) n.$$

In other words, the problem shifts toward the search for accelerated blocks in which sufficient dissipative pressure exists and, at the same time, effective control of residual memory is achieved.

Consequently, the Ruiz Castillo residual decomposition makes it possible to reformulate the study of Collatz as a problem of dynamical stability: determining whether every accelerated orbit eventually generates a block in which the dominant contraction strictly exceeds the residual contribution. This formulation does not close the problem, but offers a more precise structural path for investigating it from the interaction among number theory, symbolic dynamics, and dissipative discrete systems.

Preliminary references

References

[1] Lagarias, J. C. (1985). The $3x+1$ problem and its generalizations. *The American Mathematical Monthly, 92*(1), 3–23.

[2] Lagarias, J. C. (Ed.). (2010). *The Ultimate Challenge: The* $3x+1$ *Problem*. American Mathematical Society.

[3] Wirsching, G. J. (1998). *The Dynamical System Generated by the* $3n+1$ *Function*. Springer.

[4] Tao, T. (2019). Almost all orbits of the Collatz map attain almost bounded values. *Forum of Mathematics, Pi, 10*, e12.

[5] Ruiz Castillo, J. C. (2025). *Collatz, dynamical systems, and computational complexity*. International Journal of Science Universitam, 2(15). https://doi.org/10.5281/zenodo.14739687

[6] Ruiz Castillo, J. C. (2025). *Application of the Ontosemiotic approach and its incidence in the study of the Collatz conjecture from the sciences of complexity*. Doctoral thesis. University of San Carlos of Guatemala, Eastern University Center (CUNORI). Available at: https://biblos.usac.edu.gt/api/filesystem/v1/files/fe3d0ff7-8d1d-4772-818d-8af0336b1308

[7] Ruiz Castillo, J. C. (2026). *Residual debt, dissipative compensation, and descending subcylinders in the accelerated dynamics of the Collatz Conjecture*. Zenodo. https://doi.org/10.5281/zenodo.20320671

[8] Ruiz Castillo, J. C. (2026). *Descending subcylinders and dissipative drift in the accelerated dynamics of Collatz*. Zenodo.

[9] Ruiz Castillo, J. C. (2026). *Residual drift and average dissipative pressure in the accelerated dynamics of the Collatz Conjecture*. Zenodo.

[10] Ruiz Castillo, J. C. (2026). *Ergodic interpretation of 2-adic valuations in the accelerated dynamics of Collatz*. Zenodo.

[11] Ruiz Castillo, J. C. (2026). *Residual walks and dissipative structures in accelerated Collatz orbits*. Zenodo.

[12] Ruiz Castillo, J. C. (2026). *Balance between ternary growth and binary dissipation in the accelerated dynamics of Collatz*. Zenodo.

[13] Ruiz Castillo, J. C. (2026). *Symbolic dynamics and dissipative pressure in the Collatz Conjecture*. Zenodo.

[14] Ruiz Castillo, J. C. (2026). *2-adic valuations and residual drift in Collatz-type discrete systems*. Zenodo.

[15] Ruiz Castillo, J. C. (2026). *Ideal probabilistic model and negative drift in the accelerated dynamics of Collatz*. Zenodo.

[16] Ruiz Castillo, J. C. (2026). *Residual drift and dissipative pressure in the accelerated dynamics of the Collatz Conjecture*. Zenodo. https://doi.org/10.5281/zenodo.20481855